Om Yadao Software Engineer · AI/ML

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About

Software engineer with a quality-first, test-driven approach and hands-on experience building AI-augmented systems: Python/FastAPI REST services, multi-agent orchestration with LangGraph, LLM tool use with structured outputs, and local LLM pipelines. Comfortable across the full lifecycle (design, implementation, testing, logging, security hardening, review via pull requests) and with Docker, PostgreSQL and Redis.

Experience

AI/ML Intern — KEMURI Technology December 2025 – June 2026

- Contributed to COE AI, an AI-powered marine debris analysis system that classifies marine plastic waste into 11 categories using computer vision and multimodal AI.
- Built and maintained Python/FastAPI microservices integrating vision models and OpenAI APIs for image classification, OCR and weight estimation; designed REST endpoints for single, URL-based and batch inference, category retrieval and health monitoring.
- Engineered and refined LLM prompts for OCR and weight estimation, handling ambiguous scale displays, blurred images, decimal separators, unit variations and structured model outputs.
- Added structured response models, input validation, unit-conversion utilities and Pillow-based image validation and preprocessing for consistent, reliable inference.
- Wrote Pytest unit and integration tests (validation, preprocessing, API behavior, edge cases) and contributed to security hardening (SSRF protections), error handling, structured logging and inference performance.
- Worked in Git-based workflows with feature branches, pull requests and remote development environments to ship reliable, testable production code.

AI/ML Intern — Evoastra Ventures Pvt Ltd Oct 2025 – Nov 2025

- Worked on data-driven and predictive analytics projects under experienced mentors: model development, data preprocessing and performance optimization for client projects.

Projects

MAP — Multi-Agent AI Automation Platform

- Architected and led development of a multi-agent orchestration platform (FastAPI, LangGraph, PostgreSQL, Redis, Celery) with a React 18 and TypeScript frontend.
- Implemented asynchronous, distributed task execution through Celery and Redis queues, circuit breakers for resilience, and structured logging (Structlog) across the agent pipeline.

local-code-agent — Offline Coding Agent for Small Local LLMs

- Built an agentic development pipeline (plan, write tests, generate code, run tests, commit) in Python standard library only, running a 0.8B Qwen Q4 model through Ollama on a 2 GB GPU laptop.
- Designed a harness where code controls planning, test execution, bounded retries, rollback and commits, and the model produces small tasks in a rigid, verifiable output format; added a loop guard and retry controls.
- Delivered through 20+ reviewed pull requests with an extensive test suite, full prompt/reply transcripts for debugging, and a model benchmarking command.

Local RAG Pipeline — Ollama, FAISS & EmbeddingGemma

- Built a fully local RAG pipeline (Qwen2.5-1.5B, EmbeddingGemma) with overlapping chunking, FAISS retrieval, CrossEncoder reranking, source attribution and grounding checks.

Technical Skills

Languages: Python, C, C++, SQL; data structures and algorithms
Backend & Data: FastAPI, REST API design, Celery, Pydantic, Structlog, PostgreSQL, MySQL, Redis
AI & Agents: LangGraph, LangChain, OpenAI API, LLM tool use and structured outputs, prompt engineering, RAG, Ollama, LoRA fine-tuning (Qwen)
Engineering Practices: Pytest, unit and integration testing, TDD, Git, pull-request workflow, logging, AI coding assistants (Claude)
DevOps & Frontend: Docker, React 18, TypeScript, HTML, CSS

Education

B.E. in Artificial Intelligence and Data Science (Final year) — SRCOE, Pune 2023 – 2027

Intermediate, Maharashtra Board (60.33%) — SBES College of Science, Chh. Sambhajinagar 2021 – 2023

SSC, Maharashtra Board (84.40%) — Godavari Public School, Chh. Sambhajinagar 2020 – 2021

Languages: English, Hindi, Marathi, Japanese (speaking)